

Physical Chemistry (I)

Lecture 04

Kinetic Molecular Theory Real Gas



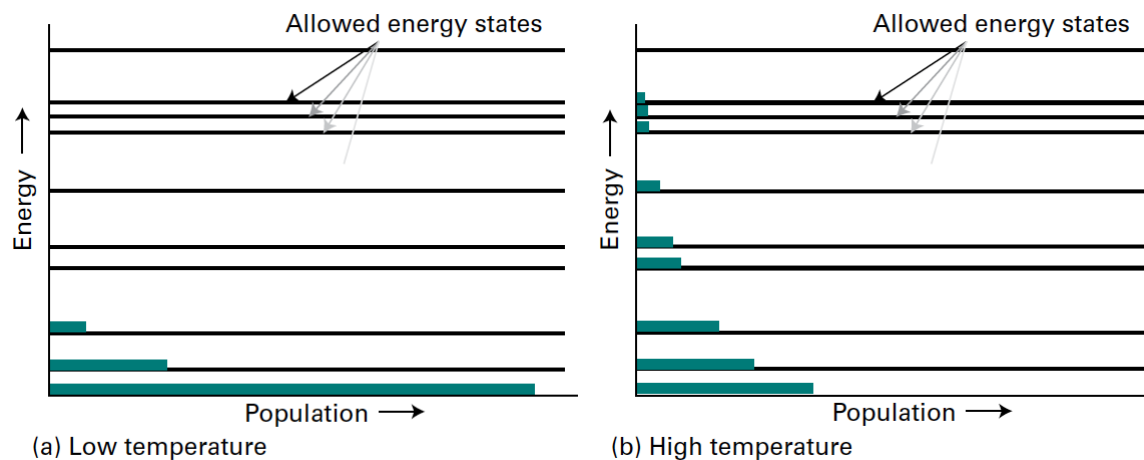
In the Last Class

- Boltzmann distribution
 - Review: probabilities
- Maxwell-Boltzmann distribution: $E \rightarrow \vec{v} \rightarrow v$
 - Characteristic speeds
- Collisions
 - Molecule-molecule collisions: chemical kinetics

Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_B T}$$

(at thermal equilibrium)



Last class

Maxwell-Boltzmann Distribution

- Velocity distribution:

$$f(v_x, v_y, v_z) = \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-M(v_x^2 + v_y^2 + v_z^2)/2RT}$$

- Speed distribution (Maxwell-Boltzmann):

$$f(v) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Last class

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M} \right)^{1/2}$$

2. Root-mean-square speed

$$v_{\text{rms}} = \left(\frac{3RT}{M} \right)^{1/2}$$

3. Most probable speed

$$v_{\text{mp}} = \left(\frac{2RT}{M} \right)^{1/2}$$



Last class

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\text{rel}} = \left(\frac{8k_{\text{B}}T}{\pi\mu} \right)^{1/2}$$

where reduced mass $\mu = m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$

Last class

Molecule-Molecule Collisions

- Collision frequency

$$z = \sigma v_{\text{rel}} \frac{n N_A}{V} = \sigma v_{\text{rel}} \frac{p}{k_B T}$$

- Mean free path

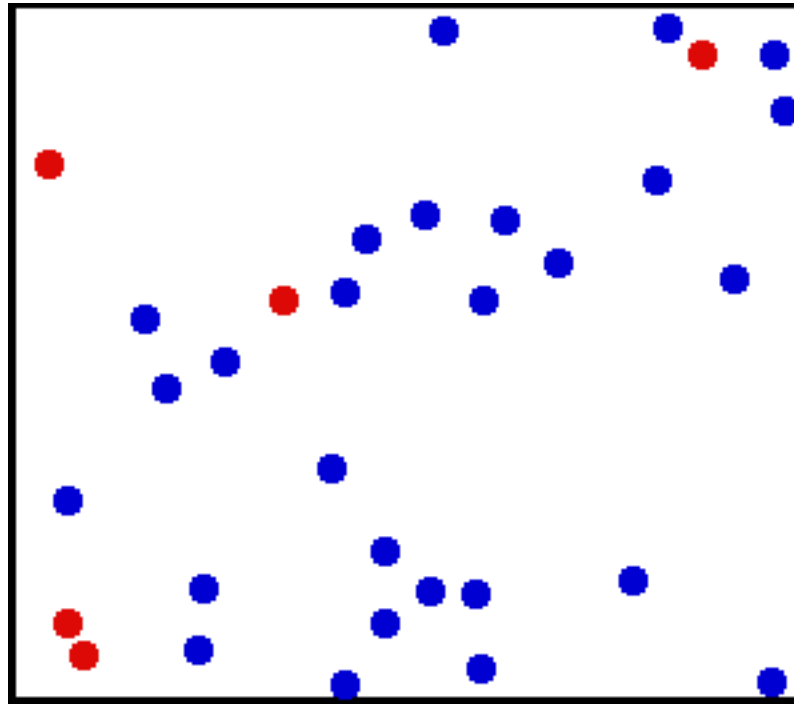
$$\lambda = \frac{v_{\text{rel}}}{z} = \frac{k_B T}{\sigma p}$$

- Collision density for a two-component system

$$Z_{AB} = \sigma v_{\text{rel}} N_A^2 \frac{n_A}{V} \frac{n_B}{V} \propto [A][B]$$



Application: Collisions



(Image: https://commons.wikimedia.org/wiki/File:Translational_motion.gif)

Consequences of Collisions

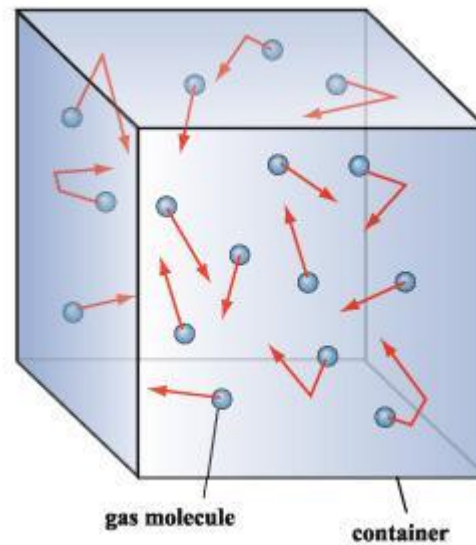
- Molecule-molecule collisions

Molecular origin of chemical reactions

- Molecule-wall collisions

Molecular origin of pressure

Molecular Origin of Pressure

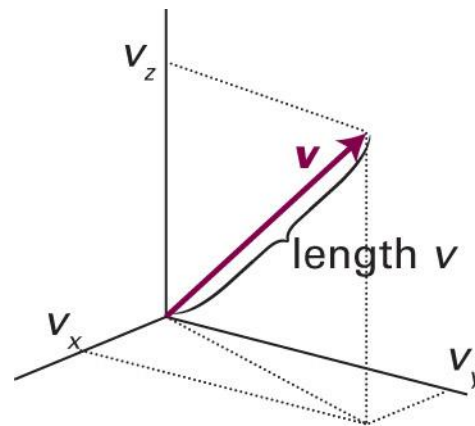


Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$



We can define linear momentum $\vec{p} = m\vec{v}$

Review

Newton's Second Law of Motion

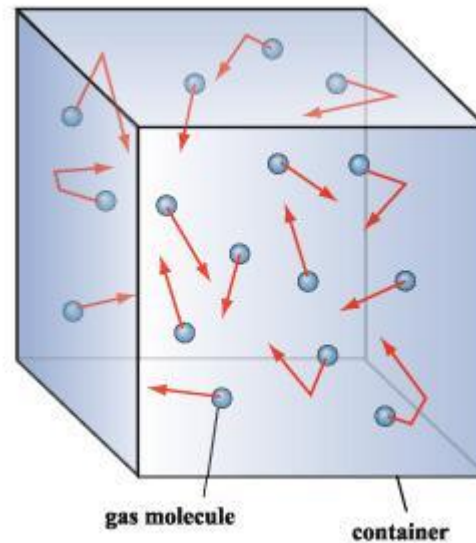
$$\text{Acceleration } \vec{a} = \frac{d\vec{v}}{dt}$$

Acceleration is determined by the force $\vec{F}$

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

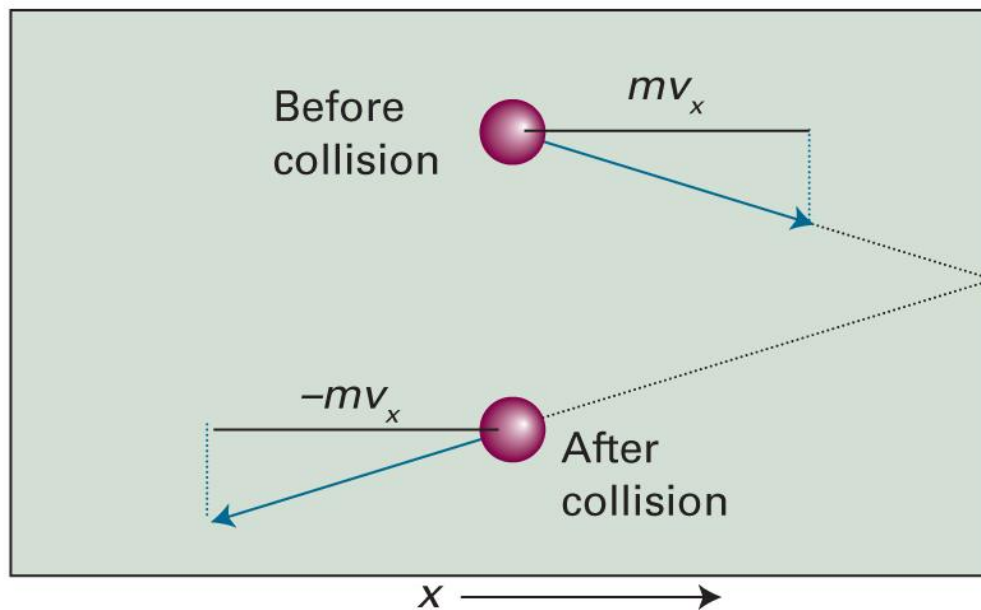
If $\vec{F} = 0$, $\frac{d\vec{p}}{dt} = 0$ (momentum conservation)

Derivation of Pressure



- Pressure $p = F/A$
- Force $\vec{F} = d\vec{p}/dt$: from collisions against the wall

The Change in Momentum



Change in momentum: $mv_x \rightarrow -mv_x$

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

Momentum Change of the Wall

- The particle has the momentum change of

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

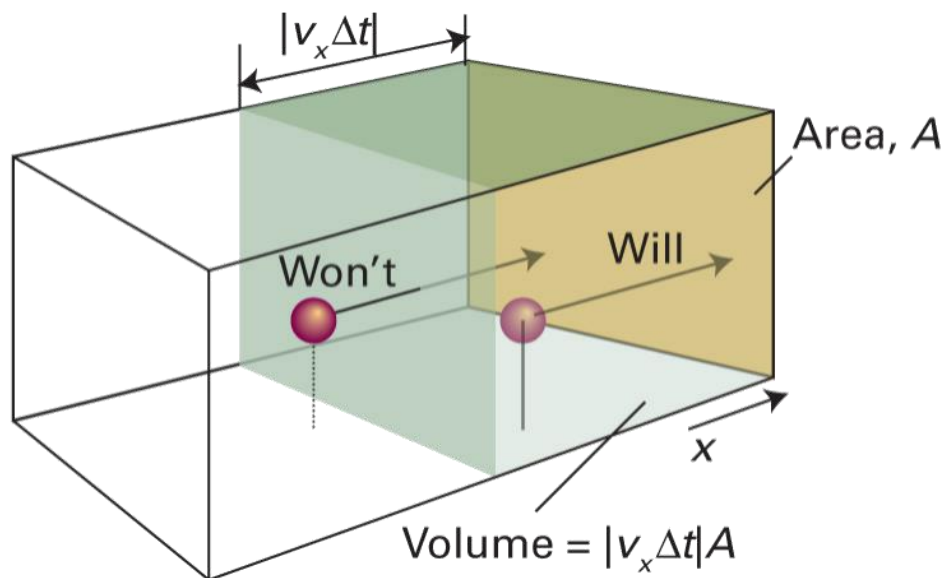
- The wall has the momentum change of

$$\Delta p_{\text{wall}} = 2mv_x$$

How Many Molecules Collide

Assume that all molecules have the same velocity.

For time Δt ,



Number of molecules = volume $\times$ density

$$= (A v_x \Delta t) \times (N_A n / V)$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1B.2)

Total Momentum Change

- For each molecule colliding the wall, the wall obtains

$$\Delta p_{\text{wall}} = 2mv_x$$

- Number of molecules = $(Av_x\Delta t) \times (N_A n/V)$
- 50% moving to the right, 50% to the left
- Total momentum change

$$\Delta p_{\text{wall, total}} = (2mv_x) \times (Av_x\Delta t) \left(\frac{N_A n}{V} \right) \times \frac{1}{2} = \frac{\overbrace{nmN_A}^M Av_x^2 \Delta t}{V}$$

Force and Pressure

- Total momentum change $\Delta p_{\text{wall,total}} = nMAv_x^2 \Delta t / V$
- Force exerted on the wall

$$F = \frac{\Delta p_{\text{wall,total}}}{\Delta t} = \frac{nMAv_x^2}{V}$$

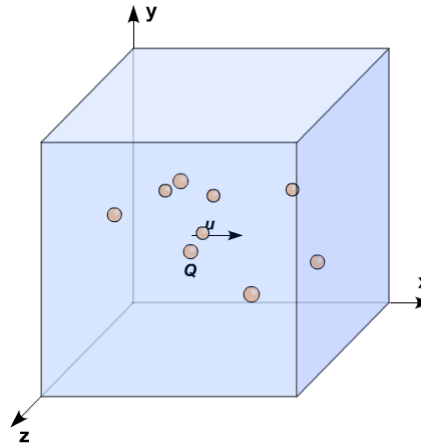
- Pressure exerted on the wall

$$p = \frac{F}{A} = \frac{nMv_x^2}{V}$$

(under the same velocity assumption)

Same Velocity Assumption

- In reality, each particle has a different velocity



- Replace the value by the mean value:

$$v_x^2 \rightarrow \langle v_x^2 \rangle$$

Pressure

$$p = \frac{nMv_x^2}{V} \rightarrow p = \frac{nM\langle v_x^2 \rangle}{V}$$

- Total velocity $v^2 = v_x^2 + v_y^2 + v_z^2$

$$\begin{aligned}\langle v^2 \rangle &= \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle \\ &= \langle v_x^2 \rangle + \langle v_x^2 \rangle + \langle v_x^2 \rangle = 3\langle v_x^2 \rangle\end{aligned}$$

- Final formula

$$p = \frac{nM\langle v^2 \rangle}{3V}$$

p - T Relationship

$$pV = \frac{1}{3} nM \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$pV = nRT$$

Internal Energy, U

“The total energy of a system”

For a perfect gas,

$$U = \sum_{i=1}^N E_{k,i} = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{1}{2} m \sum_{i=1}^N v_i^2$$

On the other hand,

$$\langle v^2 \rangle = \frac{1}{N_A n} \sum_{i=1}^N v_i^2$$

$$\sum_{i=1}^N v_i^2 = N_A n \langle v^2 \rangle$$

U of Perfect Gas

Substitution leads to

$$U = \frac{1}{2} m N_A n \langle v^2 \rangle = \frac{1}{2} M n \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$U = \frac{3}{2} nRT$$

Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each quadratic contribution
to the internal energy is $(1/2)k_{\text{B}}T$.

(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Monatomic Perfect Gas

- Potential energy = 0
- Kinetic energy: translational energy

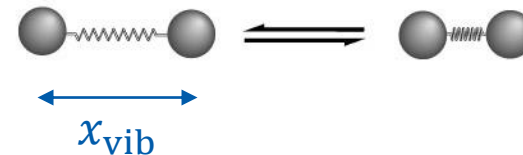
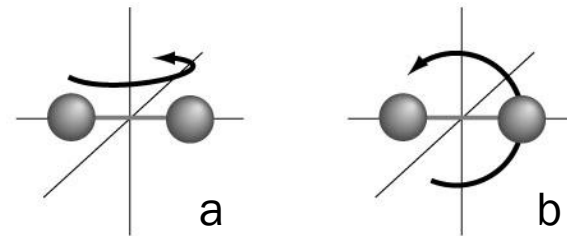
$$E_{\text{k, tr}} = \frac{p^2}{2m} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

→ 3 quadratic terms per molecule

$$U = (nN_A) \times \left(\frac{3}{2} k_B T \right) = \frac{3}{2} n N_A k_B T = \frac{3}{2} n R T$$

Diatomic Perfect Gas

- Potential energy = 0
- Kinetic energy
 - translation
 - rotation: two possible motions
- vibration: one possible motion



Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule

$$E_{k, \text{tr}} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

- Rotational energy: 2 quadratic terms per molecule

$$E_{k, \text{rot}} = \frac{p_a^2}{2m} + \frac{p_b^2}{2m}$$

- Vibrational energy: 2 quadratic terms per molecule

$$E_{k, \text{vib}} = \frac{p_{\text{vib}}^2}{2m} + \frac{1}{2} k x_{\text{vib}}^2$$

Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule
- Rotational energy: 2 quadratic terms per molecule
- Vibrational energy: 2 quadratic terms per molecule

Total: 7 quadratic terms per molecule

From the equipartition theorem,

$$U = (nN_A) \times \left(\frac{7}{2} k_B T\right) = \frac{7}{2} nN_A k_B T = \frac{7}{2} nRT$$

Polyatomic Perfect Gas

- Degree of freedom f : number of variables that may vary independently

$$f = 3a \text{ (} a: \text{ number of atoms)}$$

$$\text{CO}_2: f = 3 \times 3 = 9$$

$$\text{H}_2\text{O}: f = 3 \times 3 = 9$$

$$\text{NH}_3: f = 3 \times 4 = 12$$

Polyatomic Perfect Gas

- Linear molecules (e.g. CO₂)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 2 independent motions → 2 quadratic
 - Vibration: $(f - 5)$ independent motions → $2(f - 5)$ quadratic

$$U = (2f - 5)nRT/2$$

- Nonlinear molecules (e.g. H₂O)
 - Translation: 3 independent motions → 3 quadratic
 - Rotation: 3 independent motions → 3 quadratic
 - Vibration: $(f - 6)$ independent motions → $2(f - 6)$ quadratic

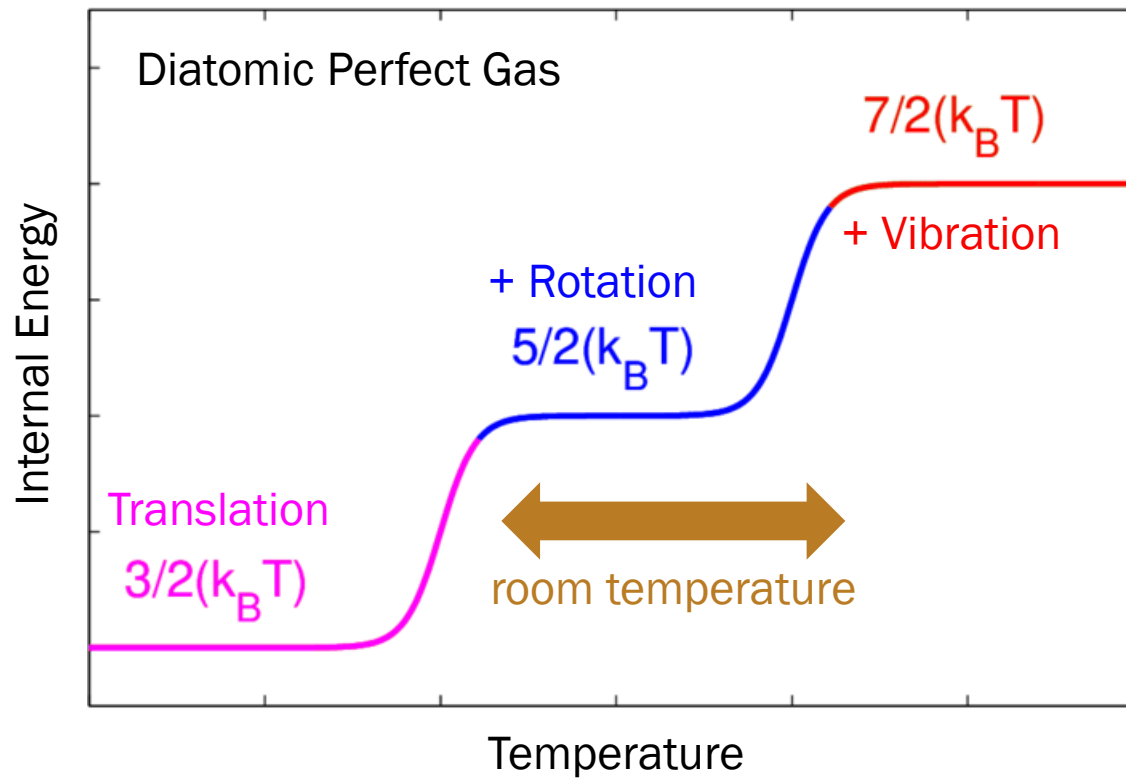
$$U = (2f - 6)nRT/2$$

U of Perfect Gas

The internal energy of a perfect gas is independent of the volume it occupies.

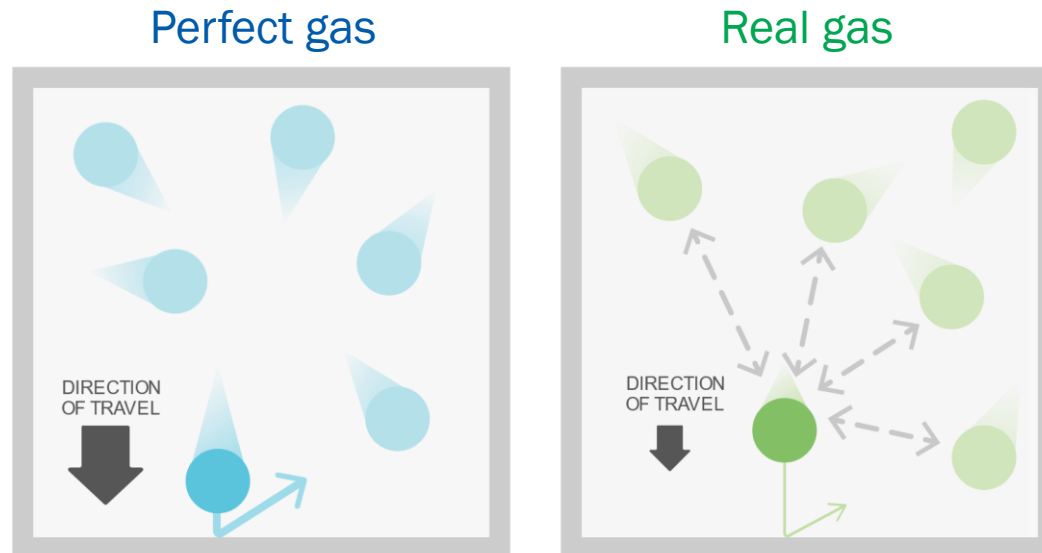
- No molecule-molecule interactions in a perfect gas
- The molecule-molecule distance has no effect on U

Excitation of Motions



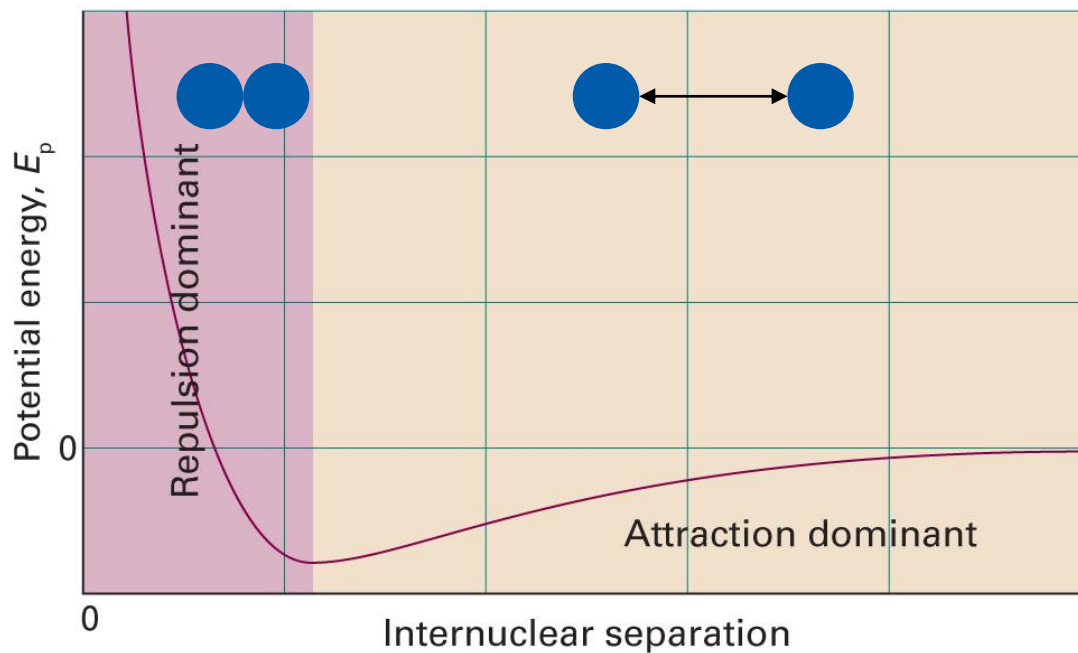
Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m/V_m^0$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^0

Since $V_m^0 = RT/p$,

$$Z = \frac{pV_m}{RT} = \frac{pV}{nRT}$$

Amplitude of Z

$$Z = V_m/V_m^0$$

- $Z > 1$:

$V_m > V_m^0 \rightarrow$ repulsion is dominant

- $Z \approx 1$:

$V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior

- $Z < 1$:

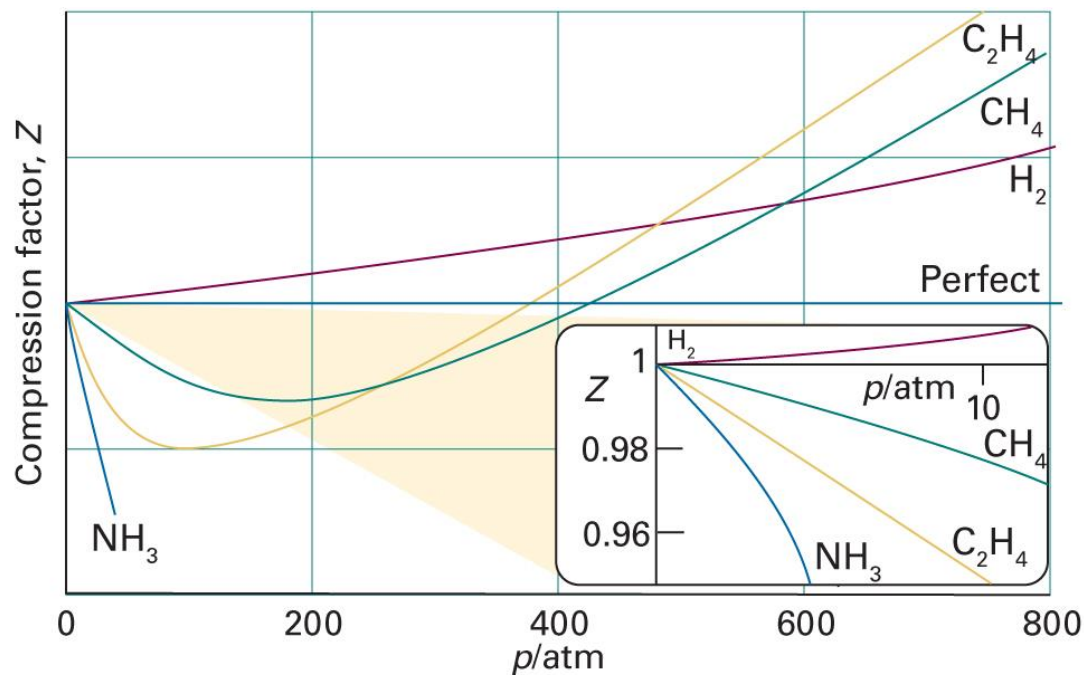
$V_m < V_m^0 \rightarrow$ attraction is dominant

Brief Illustration 1C.1

The molar volume of a perfect gas at 500 K and 100 bar is $V_m^0 = 0.416$ L mol⁻¹. The molar volume of carbon dioxide under the same conditions is $V_m = 0.366$ L mol⁻¹. What is the compression factor?

$$Z = V_m/V_m^0 = (0.366 \text{ L mol}^{-1})/(0.416 \text{ L mol}^{-1}) = 0.880$$

Experimental Values of Z



$$Z = f(p)?$$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.3)

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \dots$$

$$pV_m = RT(1 + B'p + C'p^2 + \dots)$$

The following expansion is more often used:

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Virial Coefficients

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

B : second virial coefficient

C : third virial coefficient

$\vdots$

- First virial coefficient = 1
- Virial coefficients are temperature-dependent

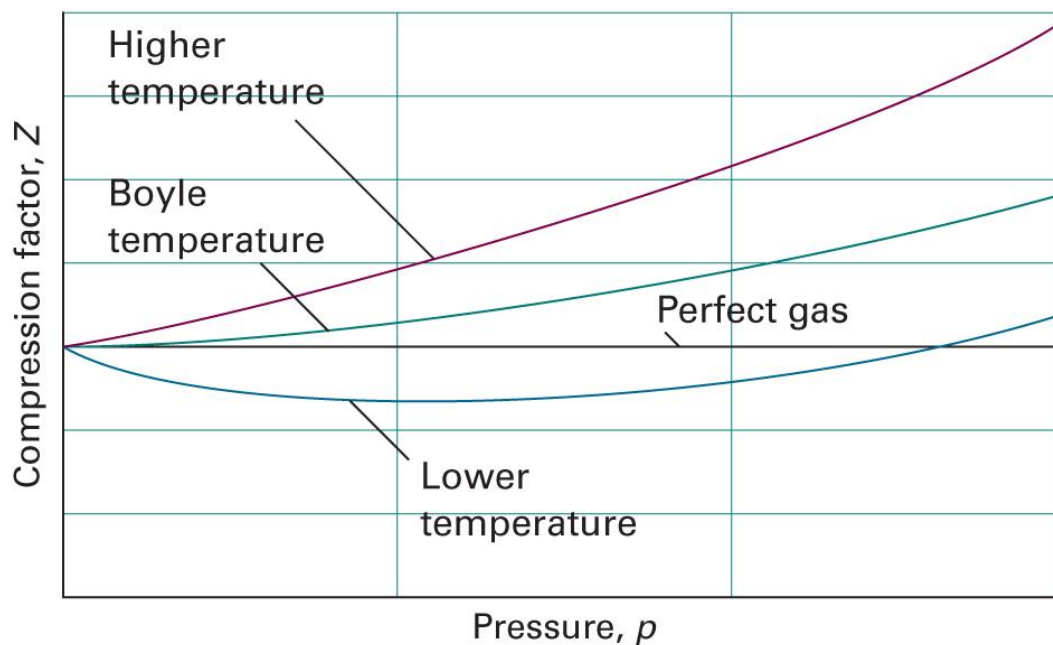
Second Virial Coefficient

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

- Generally, $B/V_m \gg C/V_m^2$
- $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
- $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
- $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Boyle Temperature T_B

Temperature where $B = 0$



Experimental Data

	$B / (\text{cm}^3 \text{ mol}^{-1})$		T_B / K
	$T = 273 \text{ K}$	$T = 600 \text{ K}$	
Ar	-21.7	11.9	411.5
CO ₂	-149.7	-12.4	714.8
N ₂	-10.5	21.7	327.2
Xe	-153.7	-19.6	768.0